NOVATEK
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Data Sheet

NT39565D
541/513/385/361 Channel Selection TFT LCD Gate Driver

V 1.0

Novatek Microelectronics Corp.

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1. General Description

The NT39565D is a dedicated 541/513/385/361 channel outputs gate driver IC for TFT-LCD panels. While a start pulse is triggered, output pins will sequentially output high-driving voltage pulses for the gate signals of the LCD panel. The NT39565D also provides shift up/down selection and cascade functions for dot expansion. The special pin location is designed for COF manufacturing.

2. Features

- 2-level Gate driver for TFT LCD panel
- 541/513/385/361 channels output selectable for normal output waveform.
- 540/512/384/360 channels output selectable for 2G, 2G+LCS output waveform.*(Refer to chip selection description)*
- Bi-directional data shift capability
- Digital supply voltage : 2.3V to 3.6V
- Maximum +40V output driving voltage
- 200KHz maximum operation frequency
- High voltage CMOS process technology
- COF package

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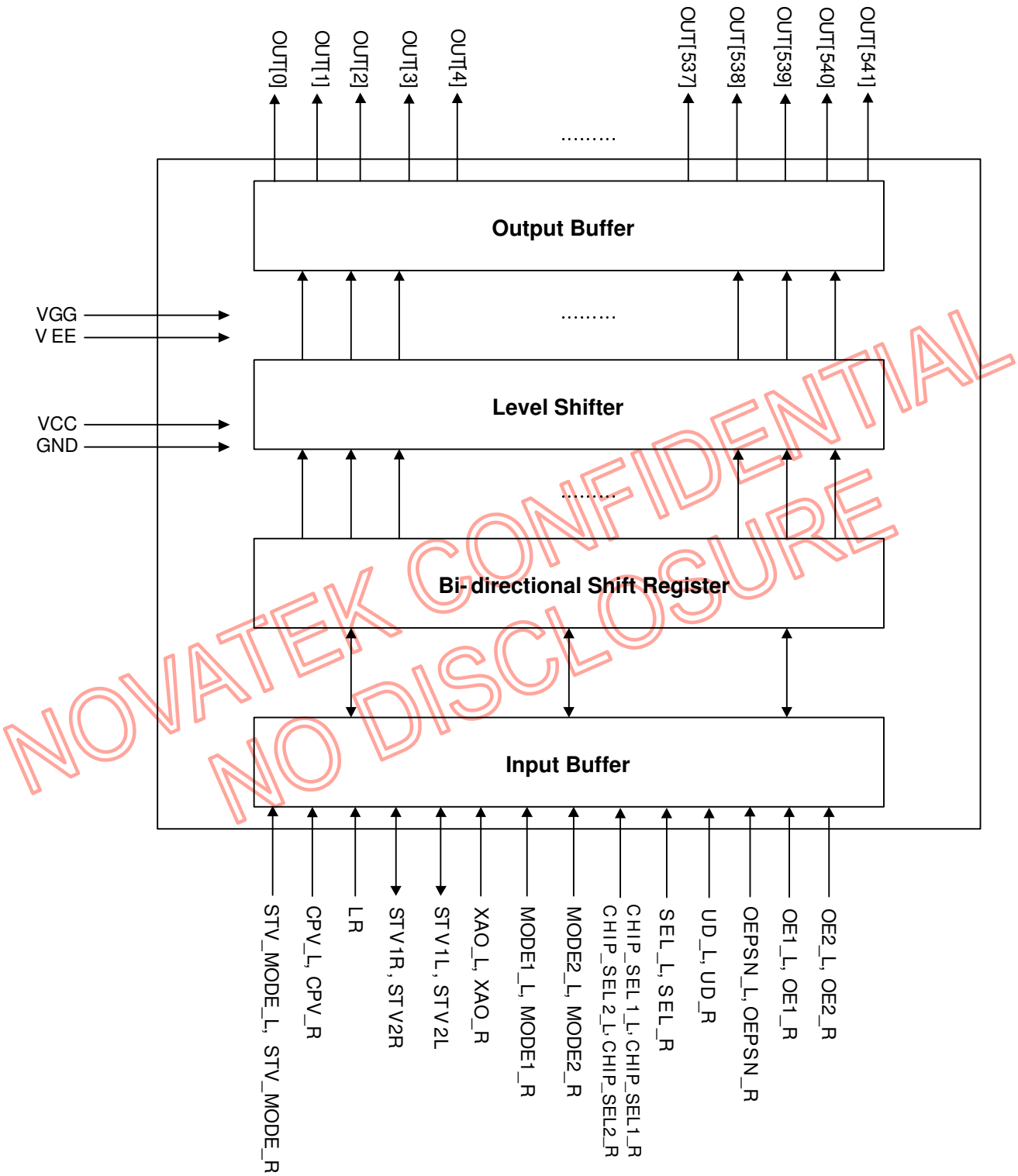
3. Block Diagram



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*Note: This functional block diagram is just for reference

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4. Pin Description

<table>
<tr><th>Designation</th><th>I/O</th><th>Descriptions</th></tr>
<tr><td>XAO_L,XAO_R</td><td>I
(Default:H)</td><td>When XAO_L / XAO_R go low, all outputs are fixed to VGG. And there is a 200k ohm pull-up resistor with this pin.
XAO_L / XAO_R signal has priority over OE1_L/R, OE2_L/R.</td></tr>
<tr><td>MODE1_L,
MODE1_R
MODE2_L,
MODE2_R</td><td>I
(Default:H)</td><td>Channel mode selection.
<table>
<tr><th>Output Channel</th><th>Disable channel (VEE level)</th><th>(CHIP_SEL2, CHIP_SEL1)</th><th>MODE2_L
MODE2_R</th><th>MODE1_L
MODE1_R</th></tr>
<tr><td>540</td><td>OUT[0], OUT[541]</td><td>(H,H)
(L,H)
(L,L)</td><td>H</td><td>H</td></tr>
<tr><td>512</td><td>OUT[0],
OUT[257]~OUT[284],
OUT[541]</td><td>(H,H)
(L,H)
(L,L)</td><td>H</td><td>L</td></tr>
<tr><td>384</td><td>OUT[0],
OUT[193]~OUT[348],
OUT[541]</td><td>(H,H)
(L,H)
(L,L)</td><td>L</td><td>H</td></tr>
<tr><td>360</td><td>OUT[0]~OUT[12],
OUT[193]~OUT[348],
OUT[529]~OUT[541]</td><td>(H,H)
(L,H)
(L,L)</td><td>L</td><td>L</td></tr>
</table>
<table>
<tr><th>Output Channel</th><th>Disable channel (VEE level)</th><th>LR</th><th>(CHIP_SEL2, CHIP_SEL1)</th><th>MODE2_L
MODE2_R</th><th>MODE1_L
MODE1_R</th></tr>
<tr><td rowspan="2">541</td><td>OUT[0]</td><td>H</td><td rowspan="2">(H,L)</td><td rowspan="2">H</td><td rowspan="2">H</td></tr>
<tr><td>OUT[541]</td><td>L</td></tr>
<tr><td rowspan="2">513</td><td>OUT[0],
OUT[257]~OUT[284],</td><td>H</td><td rowspan="2">(H,L)</td><td rowspan="2">H</td><td rowspan="2">L</td></tr>
<tr><td>OUT[257]~OUT[284],
OUT[541]</td><td>L</td></tr>
<tr><td rowspan="2">385</td><td>OUT[0],
OUT[193]~OUT[348]</td><td>H</td><td rowspan="2">(H,L)</td><td rowspan="2">L</td><td rowspan="2">H</td></tr>
<tr><td>OUT[193]~OUT[348],
OUT[541]</td><td>L</td></tr>
<tr><td rowspan="2">361</td><td>OUT[0]~OUT[12],
OUT[193]~OUT[348],
OUT[529]~OUT[540]</td><td>H</td><td rowspan="2">(H,L)</td><td rowspan="2">L</td><td rowspan="2">L</td></tr>
<tr><td>OUT[1]~OUT[12],
OUT[193]~OUT[348],
OUT[529]~OUT[541]</td><td>L</td></tr>
</table>
</td></tr>
<tr><td>CHIP_SEL1_L
CHIP_SEL1_R

CHIP_SEL2_L
CHIP_SEL2_R</td><td>I
(Default:L)</td><td>Chip selection
<table>
<tr><th>CHIP_SEL2</th><th>CHIP_SEL1</th><th>Function</th></tr>
<tr><td>H</td><td>H</td><td>2G+LCS (4 lines)</td></tr>
<tr><td>L</td><td>H</td><td>2G (Two channel pulse mode)</td></tr>
<tr><td>H</td><td>L</td><td>Normal output</td></tr>
<tr><td>L</td><td>L</td><td>2G+LCS (2 lines)</td></tr>
</table>
</td></tr>
</table>

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<table>
<tr><td>SEL_L,
SEL_R</td><td>I
(Default:H)</td><td rowspan="2">Output mode selection (For “2G+LCS (4lines)” application)
<table>
<tr><th>SEL</th><th>UD</th><th>Output mode</th><th>Output delay</th></tr>
<tr><td>H</td><td>H</td><td>One pulse mode</td><td>1CPV cycle</td></tr>
<tr><td>L</td><td>H</td><td>Jumping one pulse mode</td><td>2CPV cycle</td></tr>
<tr><td>H</td><td>L</td><td>Reverse one pulse mode</td><td>1CPV cycle</td></tr>
<tr><td>L</td><td>L</td><td>Reverse jumping one pulse mode</td><td>2CPV cycle</td></tr>
</table>
1. The “delayed” output is control by “SEL” pin. If SEL=”H”, output will delay 1CPV cycle and if SEL=”L”, output will delay 2CPV cycle. (Refer to “UD” function).
<table>
<tr><th>UD</th><th>Output mode</th></tr>
<tr><td>H</td><td>Non-reversed.</td></tr>
<tr><td>L</td><td>STV1 and STV2 start pulse are reversed.</td></tr>
</table></td></tr>
<tr><td>UD_L,
UD_R</td><td>I
(Default:H)</td></tr>
<tr><td>STV_MODE_L,
STV_MODE_R,</td><td>I
(Default:L)</td><td>STV input mode selection (For “2G+LCS (2lines)” application)
Please refer to 5-5 Operation Waveform.
<table>
<tr><th rowspan="2">STV_MODE</th><th>STV Pulse width</th><th rowspan="2">Output delay</th></tr>
<tr><th>Long (2T)</th></tr>
<tr><td>H</td><td>The first 1T (1 CPV pulse width) of the delayed outputs is masked.</td><td>1CPV cycle</td></tr>
<tr><td>L</td><td>No masked for the delayed outputs.</td><td>1CPV cycle</td></tr>
</table></td></tr>
<tr><td>OE1_L,OE1_R,
OE2_L,OE2_R</td><td>I</td><td>Output enables control.
The OE1_L, OE1_R & OE2_L, OE2_R signals control the duty cycle width of output channel.
When OE goes high, output will pulled to low level output (VEE).
When 541/540CH, (CHIP_SEL1&2 = H/L)
OE1_L/R is used to control :
OUT[1],OUT[3],OUT[5],……,OUT537],OUT[539] ,OUT[541]
OE2_L/R is used to control :
OUT[0],OUT[2],OUT[4],OUT[6],……OUT[538],OUT[540]
When 513/512CH, (CHIP_SEL1&2 = H/L)
OE1_L/R is used to control :
OUT[1],OUT[3],OUT[5],…OUT[255],OUT[285],…,OUT[537],OUT[539] ,OUT[541]
OE2_L/R is used to control :
OUT[0],OUT[2],OUT[4],OUT[6],…OUT[256],OUT[286]….,OUT[538],OUT[540]
When 385/384CH, (CHIP_SEL1&2 = H/L)
OE1_L/R is used to control :
OUT[1],OUT[3],OUT[5],…OUT[191],OUT[349],…,OUT[537],OUT[539] ,OUT[541]
OE2_L/R is used to control :
OUT[0],OUT[2],OUT[4],OUT[6],…OUT[192],OUT[350],…,OUT[538],OUT[540]
When 361/360CH, (CHIP_SEL1&2 = H/L)
OE1_L/R is used to control :
OUT[13],OUT[15],OUT[17],…OUT[191],OUT[349],…OUT[527],OUT[541]
OE2_L/R is used to control :
OUT[0],OUT[14],OUT[16],OUT[18],…OUT[192],OUT[350],…OUT[528]</td></tr>
</table>

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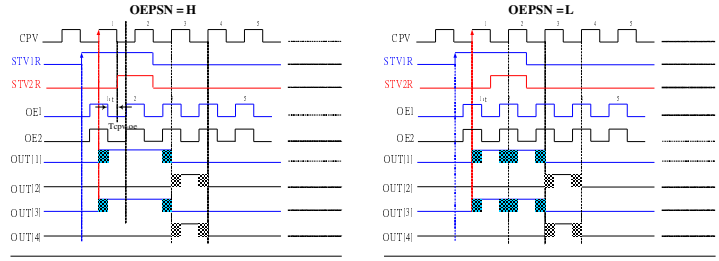
<table>
<tr><td>OEPSN_L,
OEPSN_R</td><td>I
(Default:L)</td><td>Select OE mask function.
<table>
<tr><th>OEPSN</th><th>Output Channel</th></tr>
<tr><td>H</td><td>OE1 and OE2 is activated only on the rising edge and falling edge of OUT[n].</td></tr>
<tr><td>L</td><td>Normal OE function.</td></tr>
</table>


1.Tcpv-oe > 0</td></tr>
<tr><td>LR</td><td>I</td><td>Shift up or down control. Please refer to STV1R, STV2R, STV1L, STV2L function description.</td></tr>
<tr><td>CPV_L, CPV_R</td><td>I</td><td>Shift clock. Clock signal for internal shift register.</td></tr>
<tr><td>STV1R
STV2R
STV1L
STV2L</td><td>I/O</td><td>Start pulse input/output.
These pins are used to input and output shift data. These pins are switched between input and output by setting the LR pin as below table.

1. Single STV input (Only in Normal mode & Two channel pulse mode (2G) function)
<table>
<tr><td>LR</td><td></td></tr>
<tr><td>H</td><td>STV1R is used for start pulse input.</td></tr>
<tr><td>L</td><td>STV1L is used for start pulse output.</td></tr>
</table>
----STV2R and STV2L are floating.

2. Dual STV input (Only in 2G+LCS(2 lines) and 2G+LCS(4 lines) function)
<table>
<tr><td>LR</td><td></td></tr>
<tr><td>H</td><td>STV1R, STV2R is used for start pulse input.
STV1L, STV2L is used for start pulse output.</td></tr>
<tr><td>L</td><td>STV1L, STV2L is used for start pulse input.
STV1R, STV2R is used for start pulse output.</td></tr>
</table>
1. When set for input
These pins are used as the start pulse to the shift registers of the gate driver.
The data is latched into the shift registers at the rising edge of CPV_L/R.

2. When set for output
When two or more gate drivers are cascaded, these pins output the start pulse to the next stage.</td></tr>
<tr><td>VGG</td><td>-</td><td>Positive power supply for OUT[0]~ OUT[541] outputs</td></tr>
<tr><td>VEE</td><td>-</td><td>Negative power supply for OUT[0] ~ OUT[541] outputs</td></tr>
<tr><td>VCC</td><td>-</td><td>Power supply for digital circuit.</td></tr>
<tr><td>GND</td><td>-</td><td>Ground pin</td></tr>
<tr><td>OUT[0]~OUT[541]</td><td>O</td><td>For driving LCD's gate signals, the output amplitude of these pins is VGG – VEE. Output timing of these signals is synchronous with the rising edge of shift clock.</td></tr>
</table>

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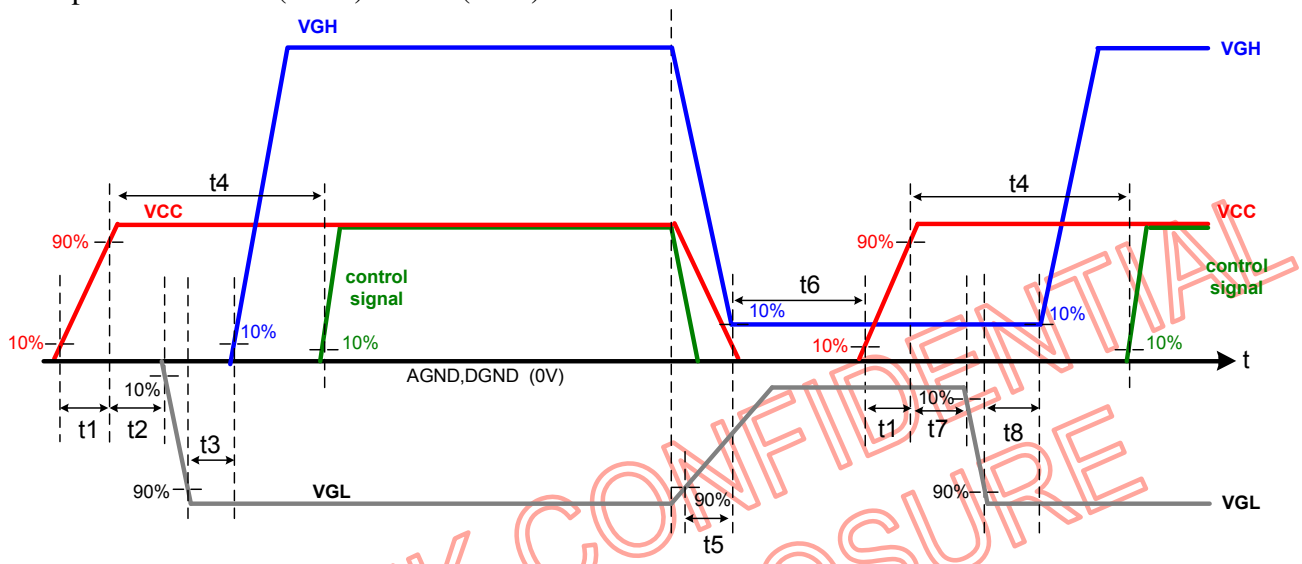
5. Functional Description

5-1. Power On / off sequence

To prevent the device damage from latch up, the power ON/OFF sequence shown below must be followed.

When power on: VCC→VGL(VEE)→VGH(VGG)
When power off: VGH(VGG)→VGL(VEE)→VCC



Timing	Min.	Typ.	Max.	unit	note
t1	0.5		10	ms	Refer to VESA timing
t2	0			ms	
t3	0			ms	
t4	0			ms	
t5	0			ms	
t6	0			ms	
t7	0			ms	
t8	0			ms	

(Display image may be abnormal before analog voltage reaches it's target value)

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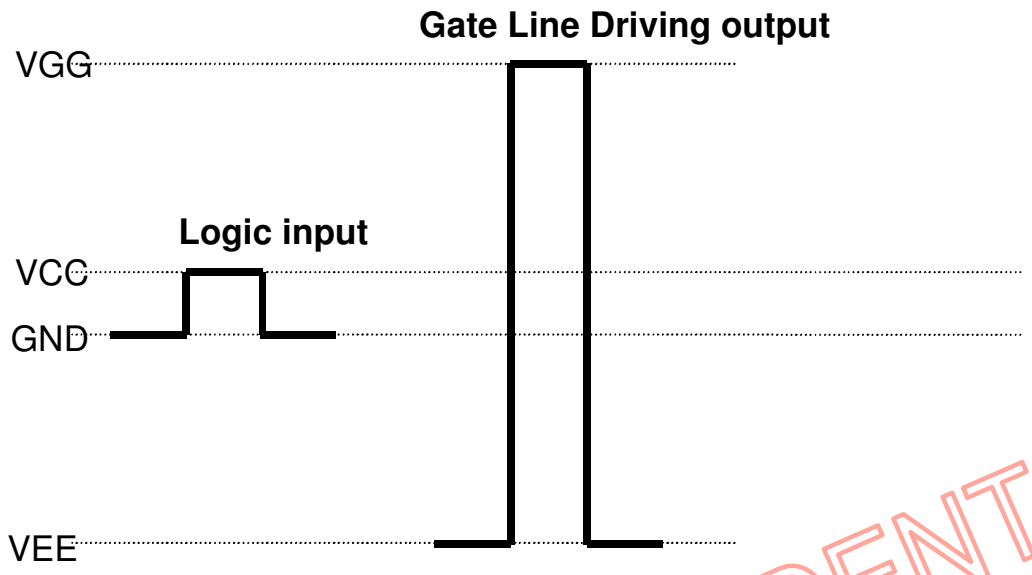
5-2. Power level:



Note: For the input signals: **CPV_L/R, XAO_L/R, MODE1_L/R, MODE2_L/R, CHIP_SEL1_L/R, CHIP_SEL2_L/R, SEL_L/R, UD_L/R, OE1_L/R, OE2_L/R, OEPSN_L/R, STV1R, STV2R, STV1L, STV2L, STV_MODE_L/R & LR**

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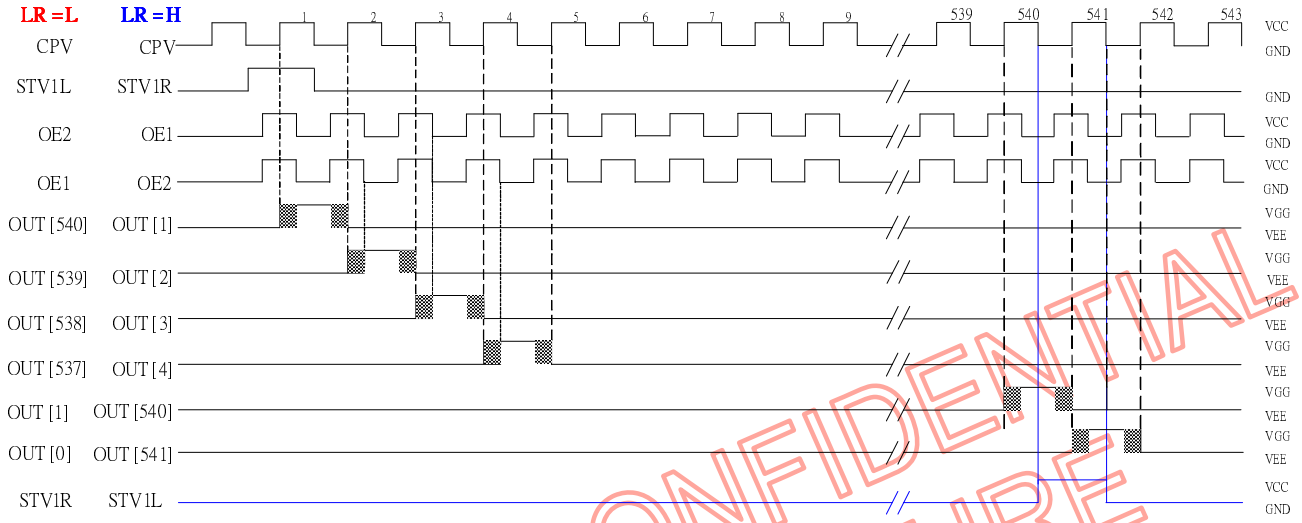
5-3. Operation Waveform (**Normal output**)

1. Short STV input, Output channel = 541

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	L	H / L	H / L	L

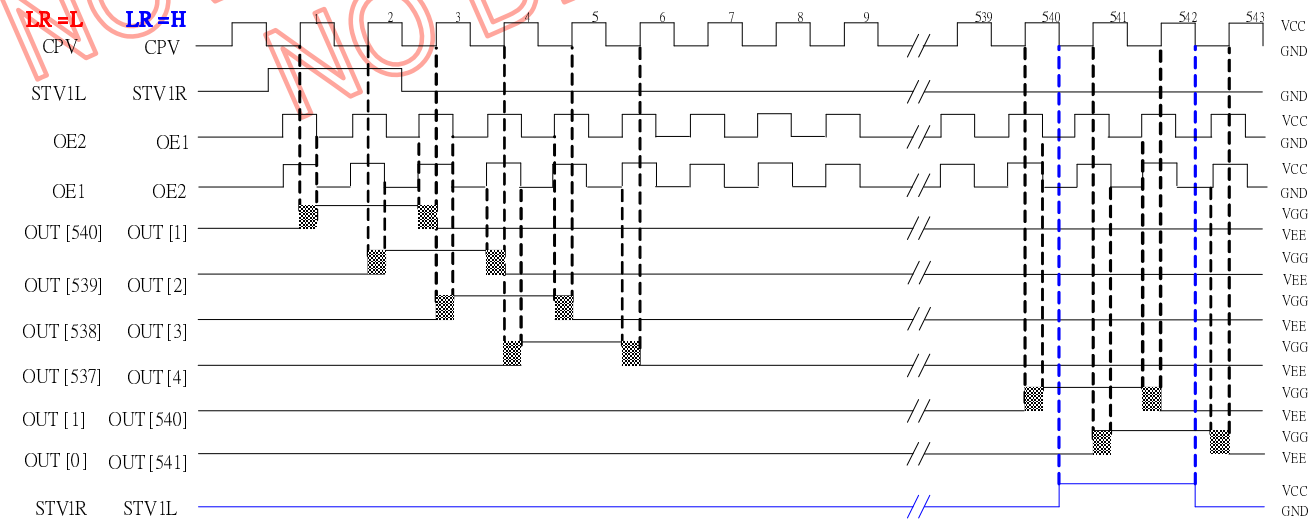
("H" level =VCC, "L" level =GND)



2. Long STV input, Output channel = 541

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	L	H / L	H / L	H

("H" level =VCC, "L" level =GND)



Note : "▩" Output masked by OE

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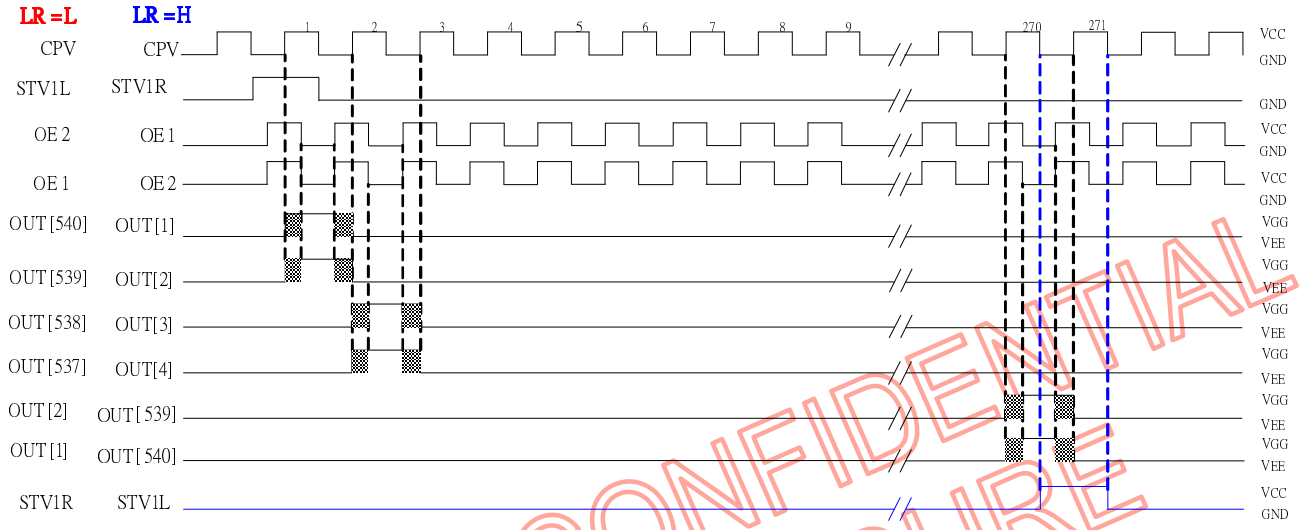
5-4. Operation Waveform (**Two Channel Pulse mode**)

1. Short STV input, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	L	H	H / L	H / L	L

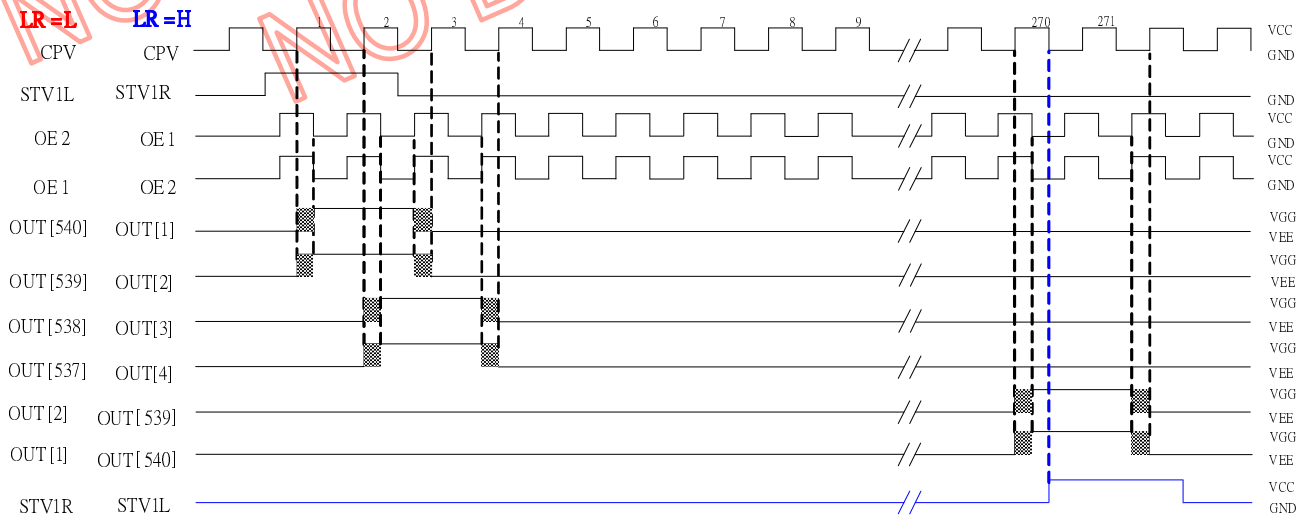
("H" level =VCC, "L" level =GND)



2. Long STV input, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	L	H	H / L	H / L	H

("H" level =VCC, "L" level =GND)



Note : "▩" Output masked by OE

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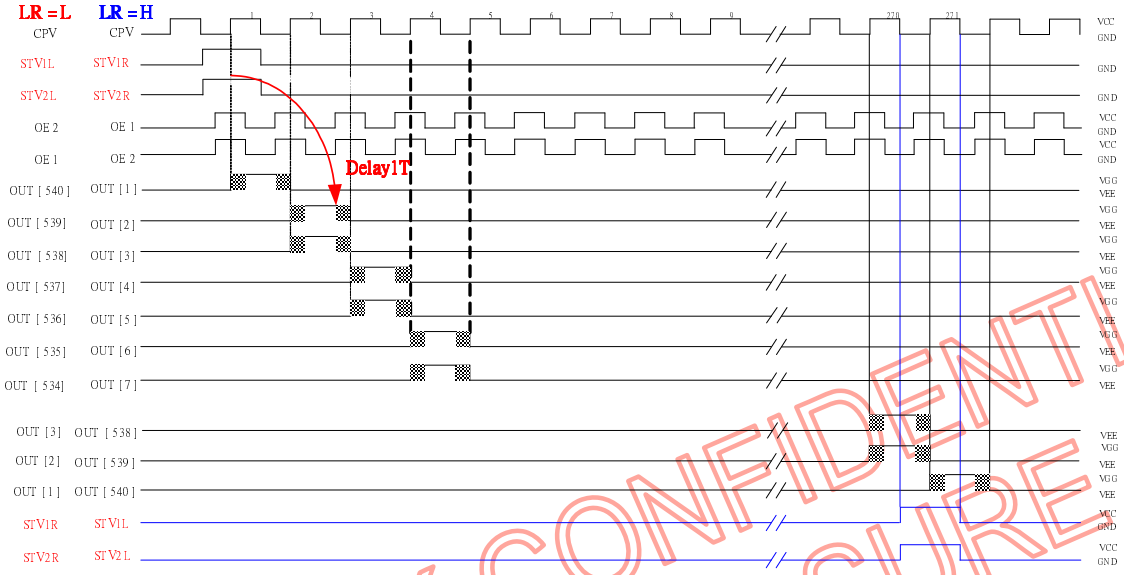
5-5. Operation Waveform (**2G+LCS 2 lines**)

1. Short STV input, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	H	H	L	L

("H" level =VCC, "L" level =GND)

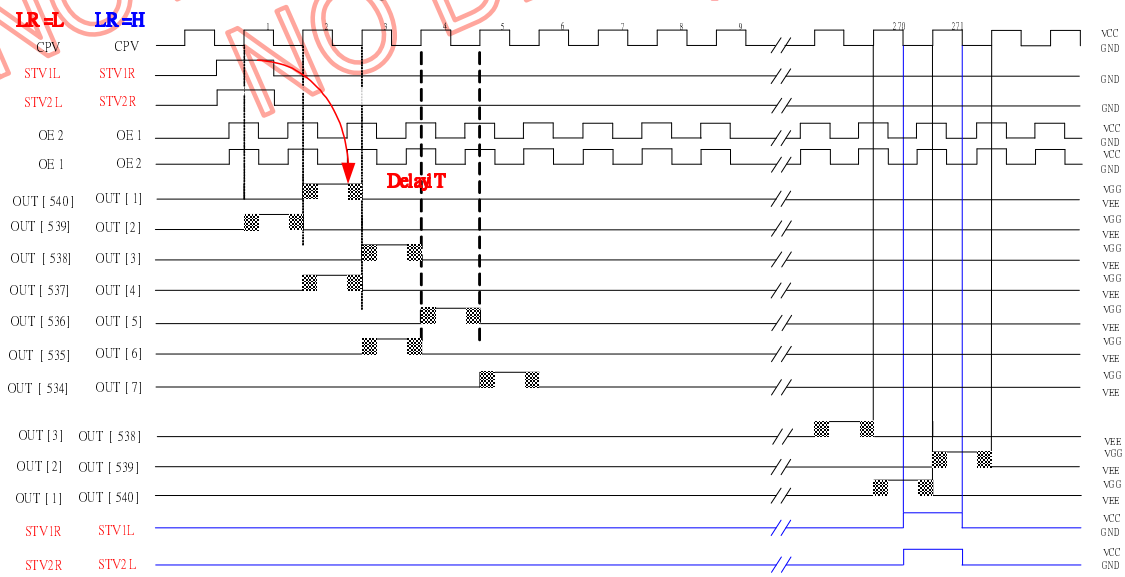
(a) Identical STV1 & STV2 timing **(UD =H with 1st channel output first)**



LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	H	L	L	L

("H" level =VCC, "L" level =GND)

(b) Identical STV1 & STV2 timing **(UD =L with 2nd channel output first)**



Note : "▦" Output masked by OE

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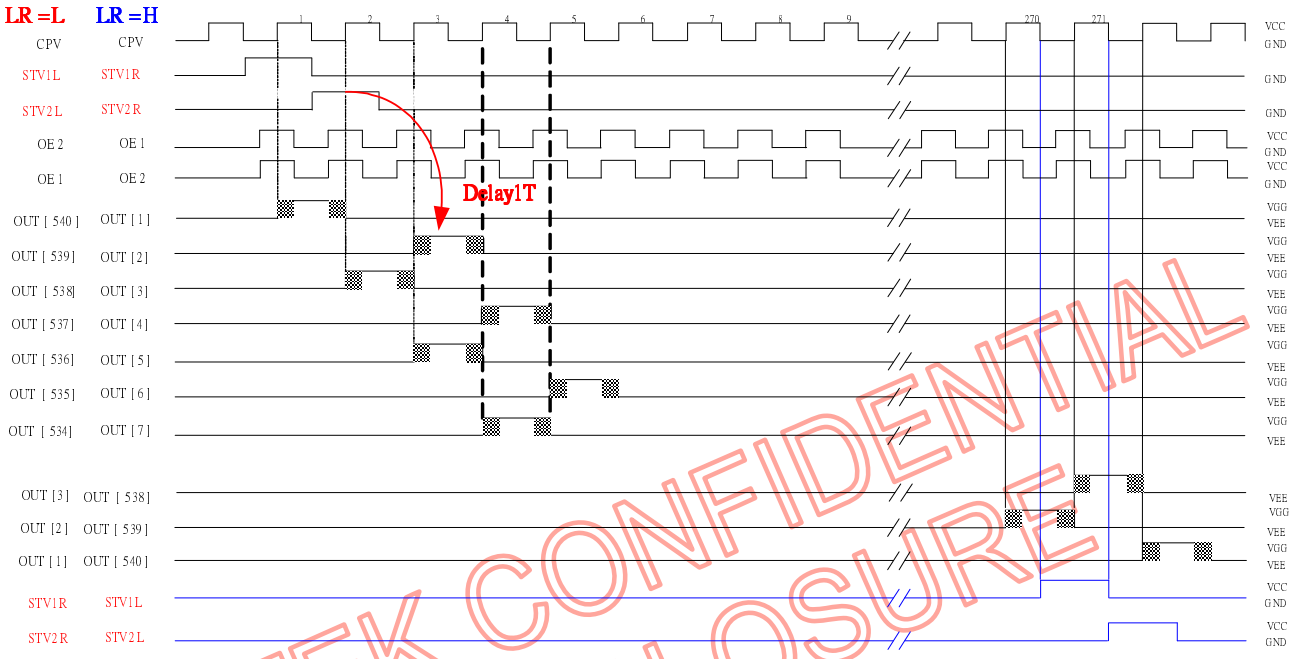
LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	H	H	L	L

("H" level =VCC, "L" level =GND)

(c) Delay 1CLK timing between STV1 & STV2



Note : "▩" Output masked by OE

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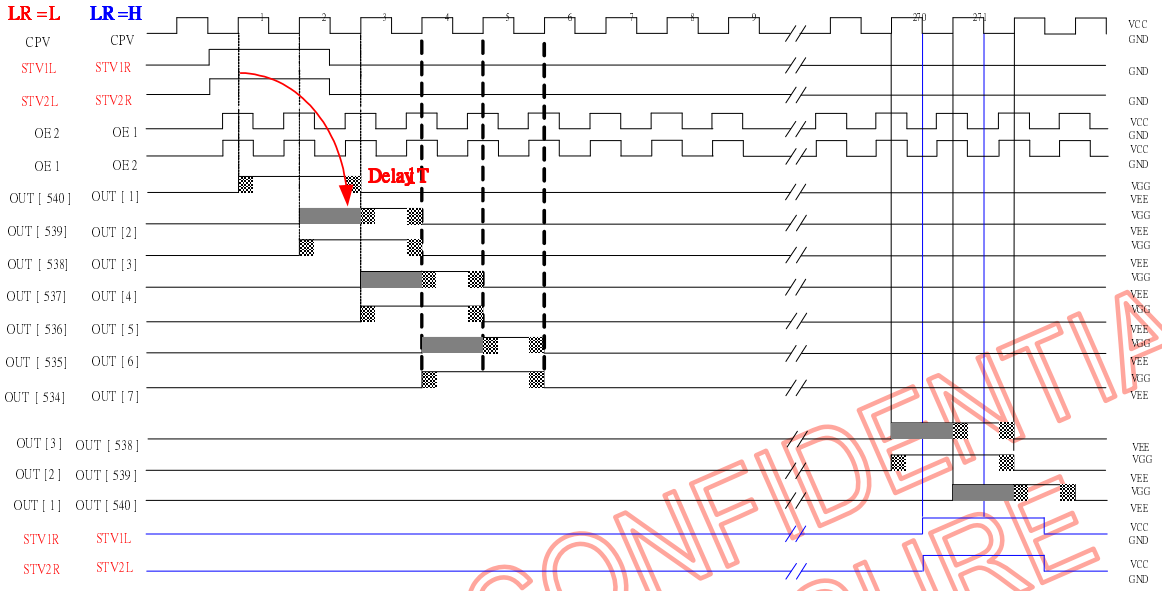
2. Long STV input, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	H	H	H	H

("H" level =VCC, "L" level =GND)

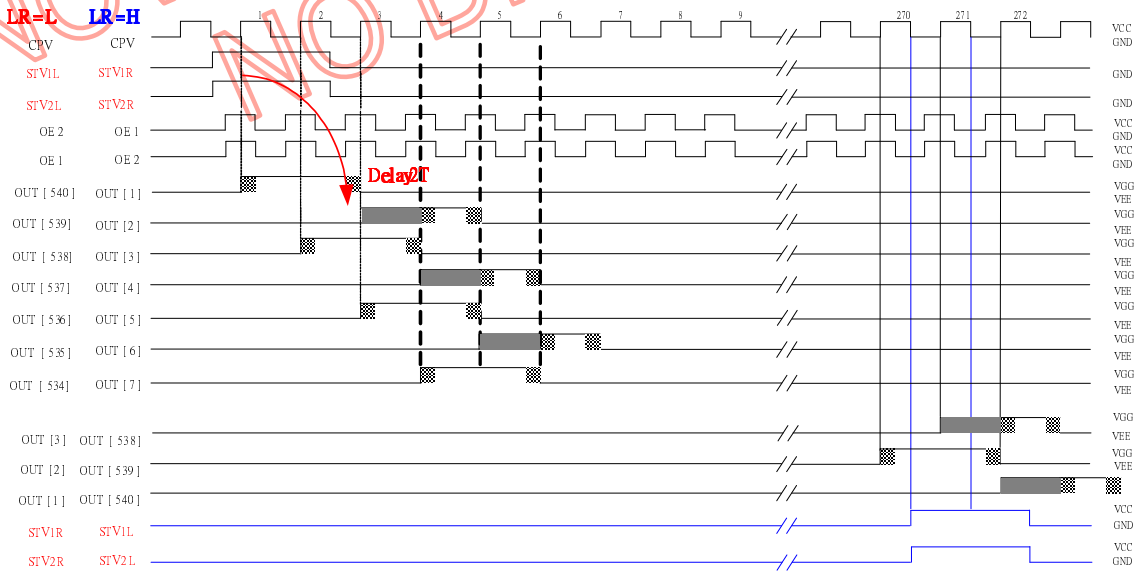
(a) Identical STV1 & STV2 timing



LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	L	H	H	H

("H" level =VCC, "L" level =GND)

(b) Identical STV1 & STV2 timing



Note : "▦" Output masked by OE ; "■" Output masked STV_MODE

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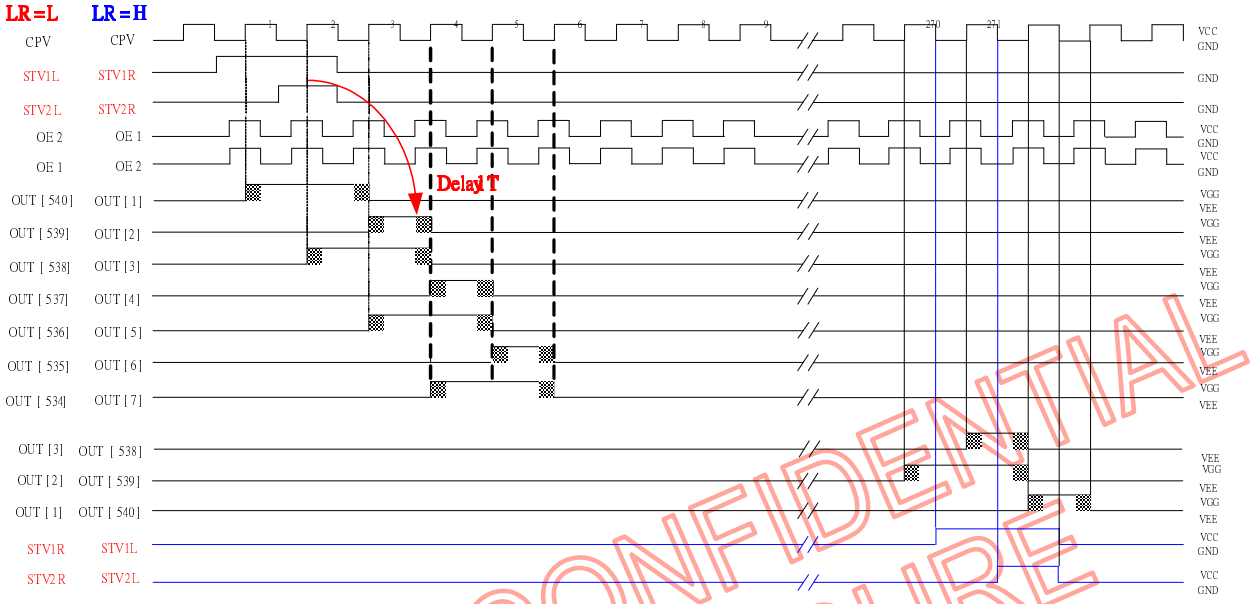
LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	H	H	H	L

("H" level =VCC, "L" level =GND)

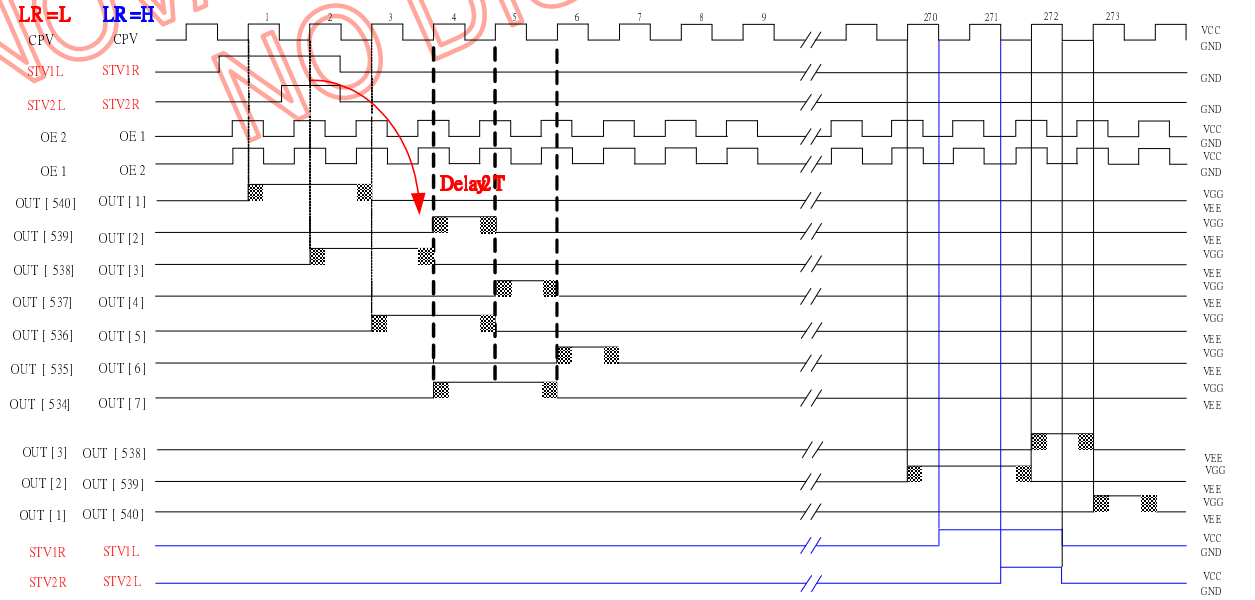
(c) Delay 1CLK timing between STV1 & STV2



LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN	STV_MODE
H/L	H	H	L	L	L	H	H	L

("H" level =VCC, "L" level =GND)

(d) Delay 1CLK timing between STV1 & STV2



Note : "▩" Output masked by OE

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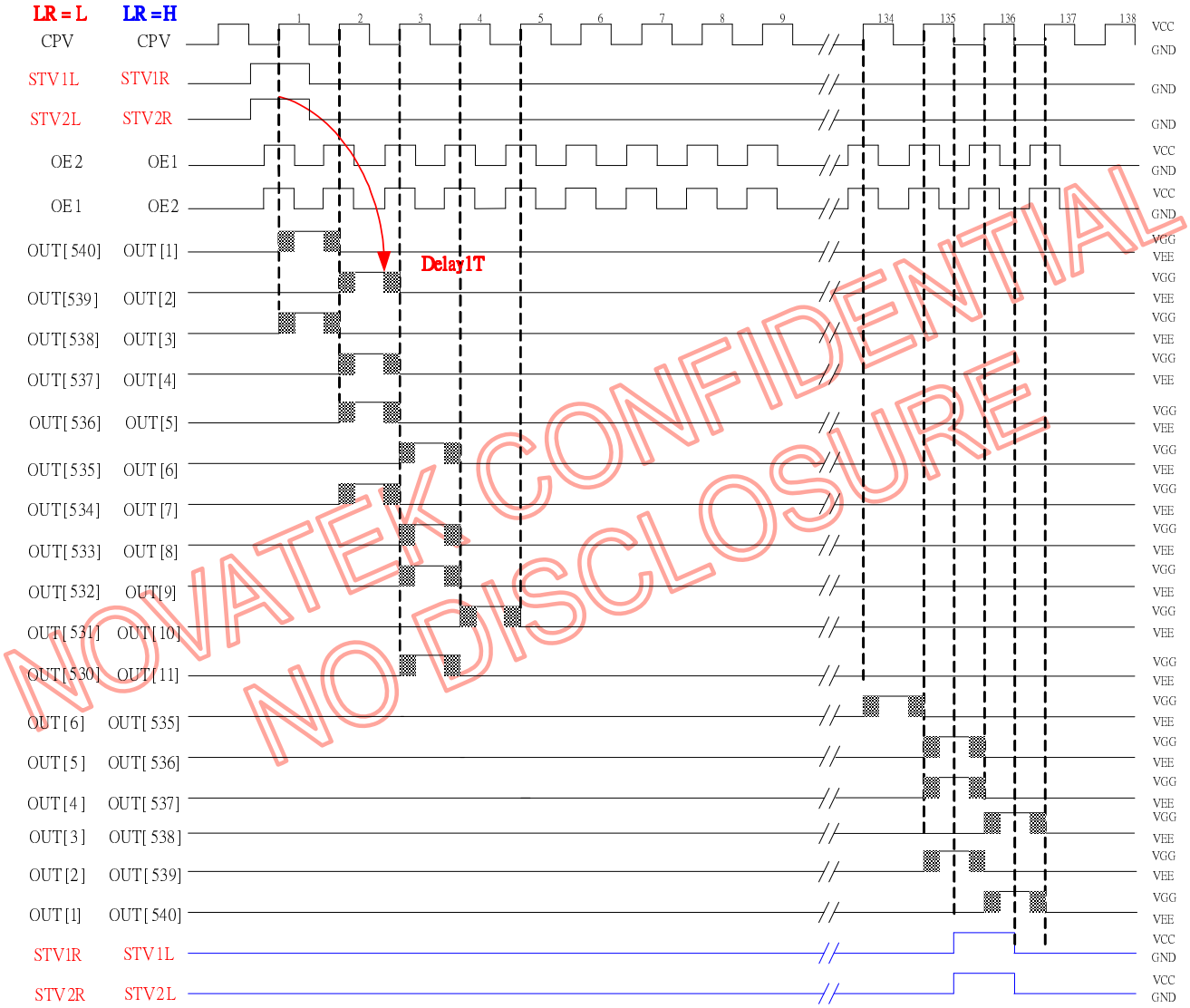
5-6. Operation Waveform (**2G+LCS 4 lines**)

1. **One pulse mode**, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	H	H	L

("H" level =VCC, "L" level =GND)



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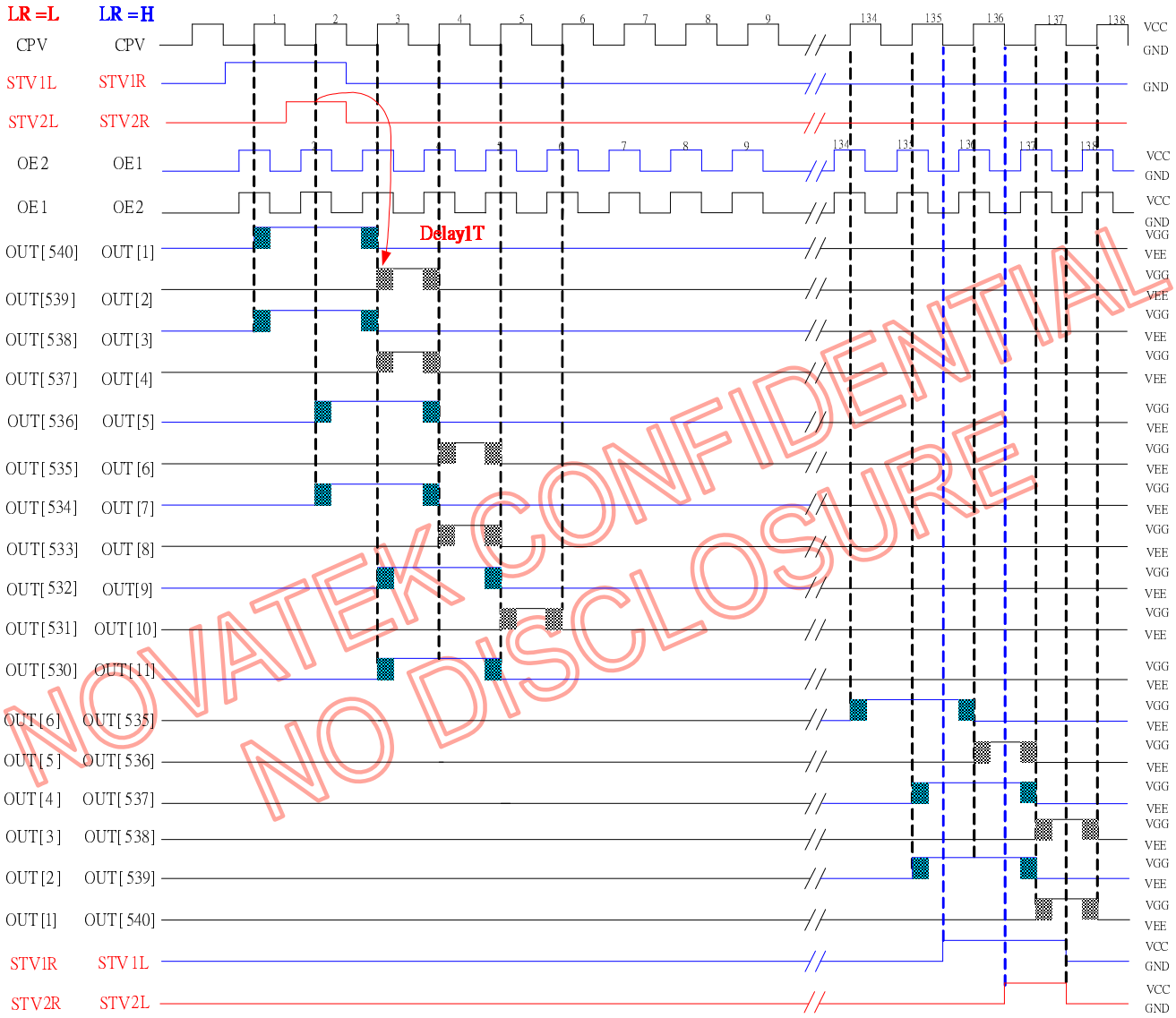
2. One pulse mode (Long STV), Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	H	H	H

("H" level =VCC, "L" level =GND)



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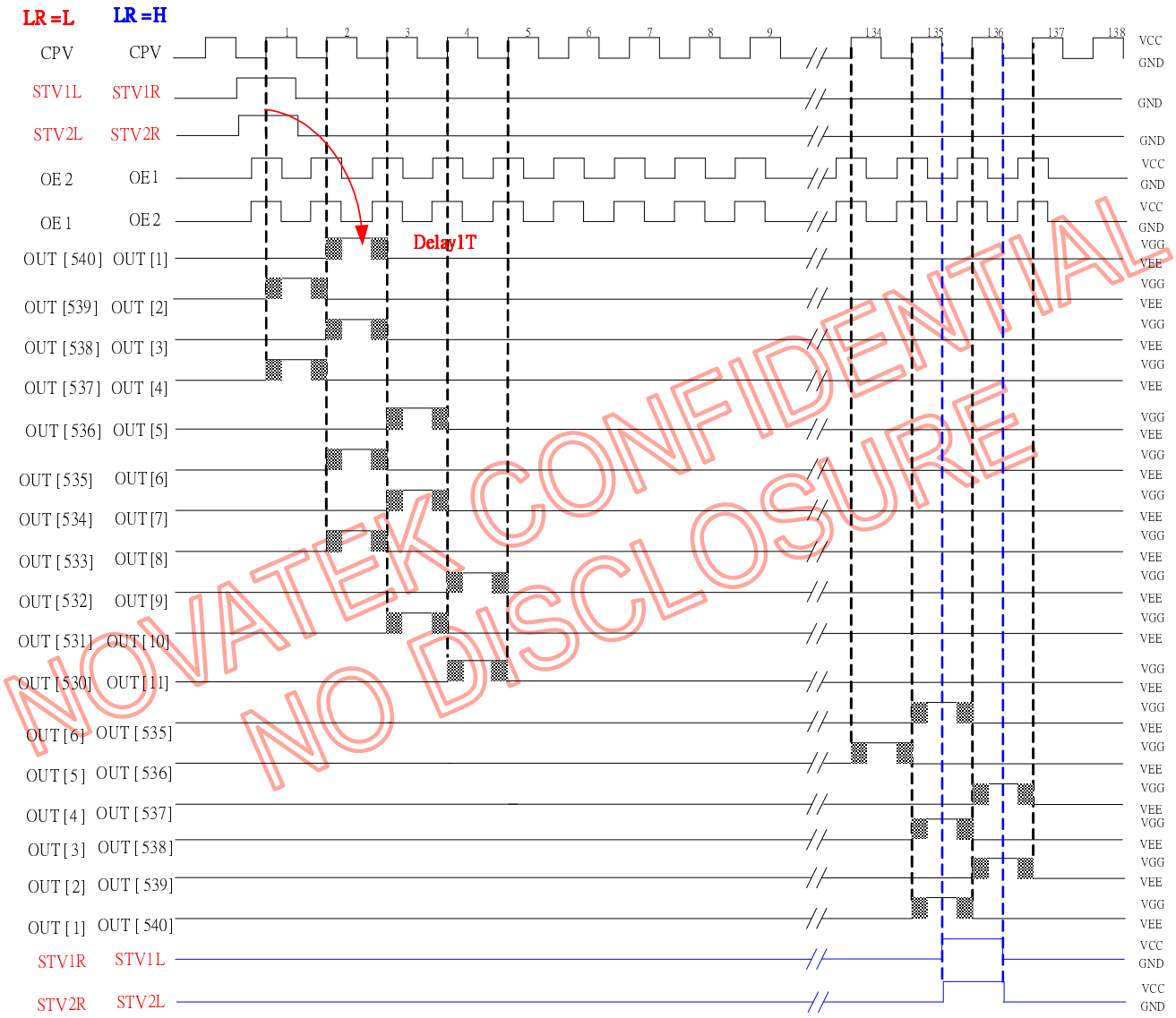
3. **Reverse One pulse mode**, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	H	L	L

("H" level =VCC, "L" level =GND)



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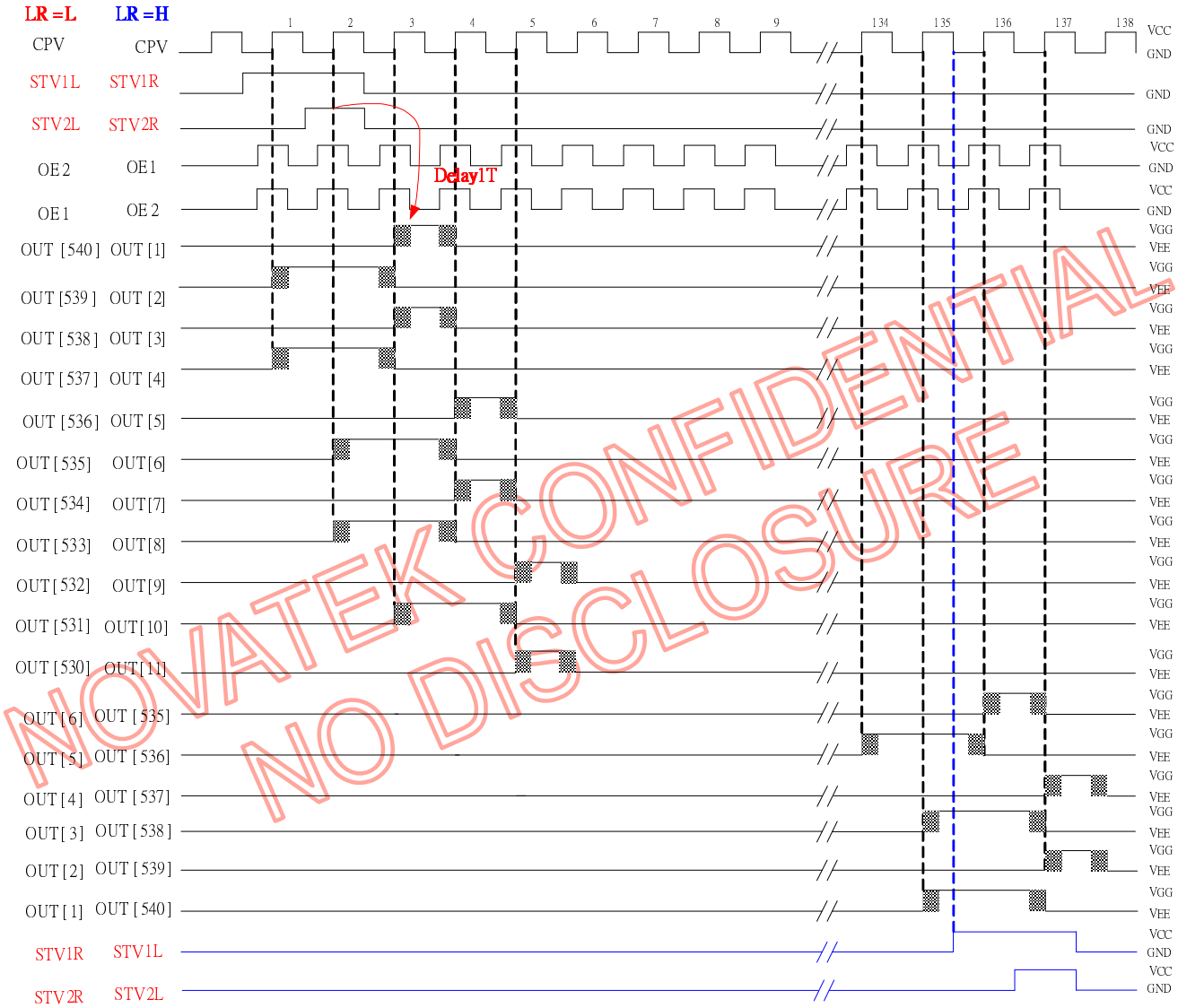
4. **Reverse One pulse mode(Long STV)**,Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	H	L	H

("H" level =VCC, "L" level =GND)



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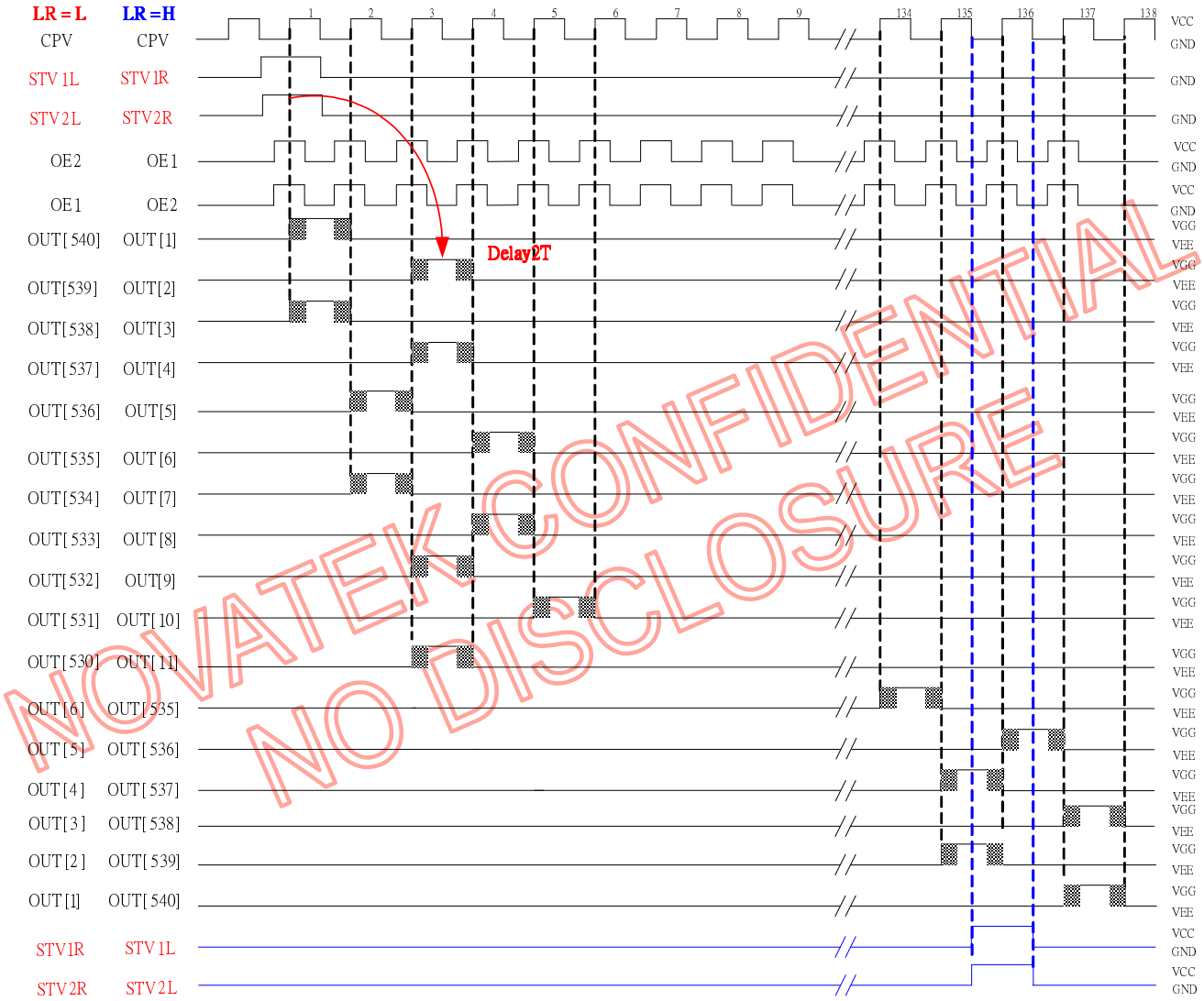
5. **Jumping One pulse mode**, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	L	H	L

("H" level =VCC, "L" level =GND)



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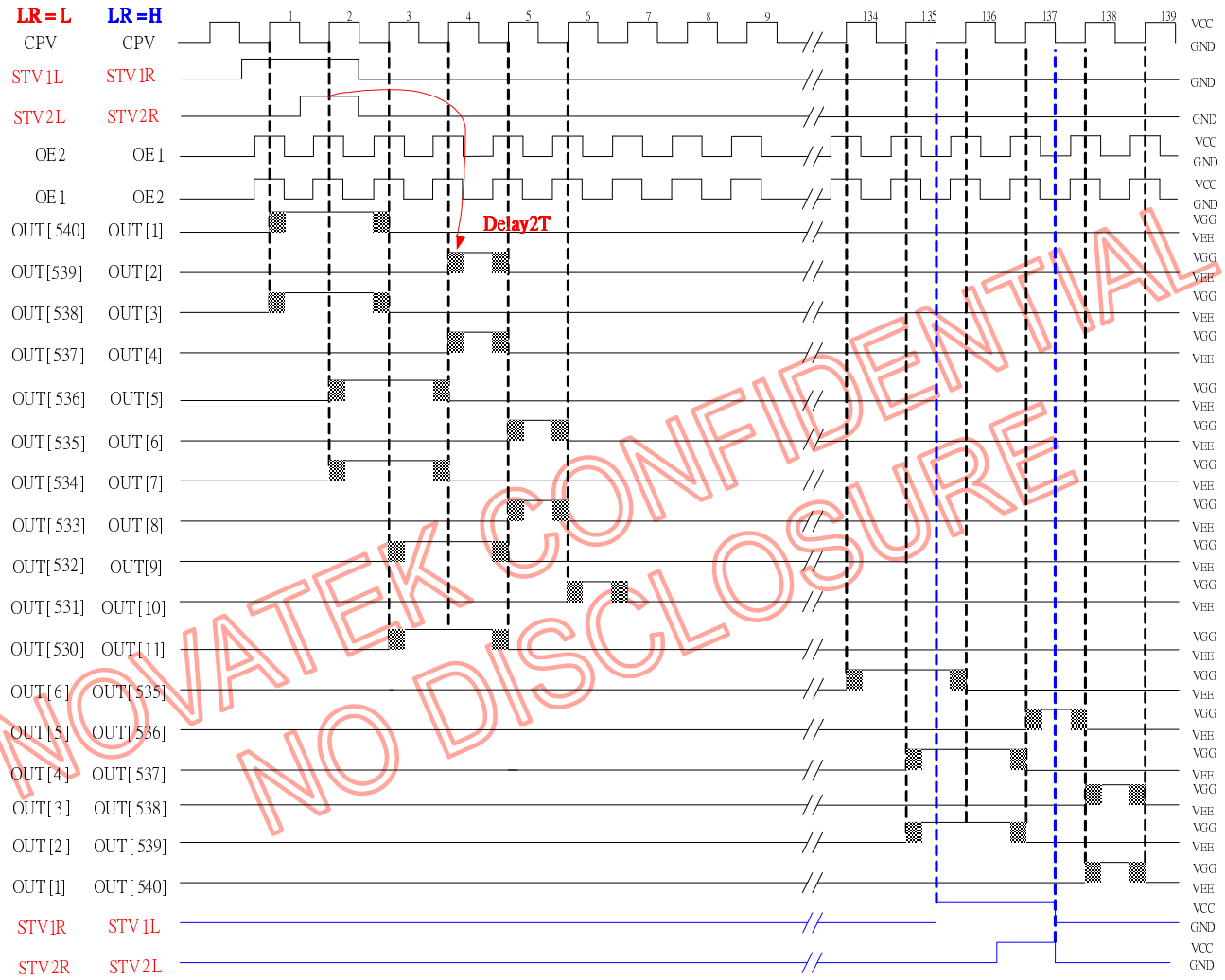
6. Jumping One pulse mode (Long STV), Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	L	H	H

("H" level =VCC, "L" level =GND)



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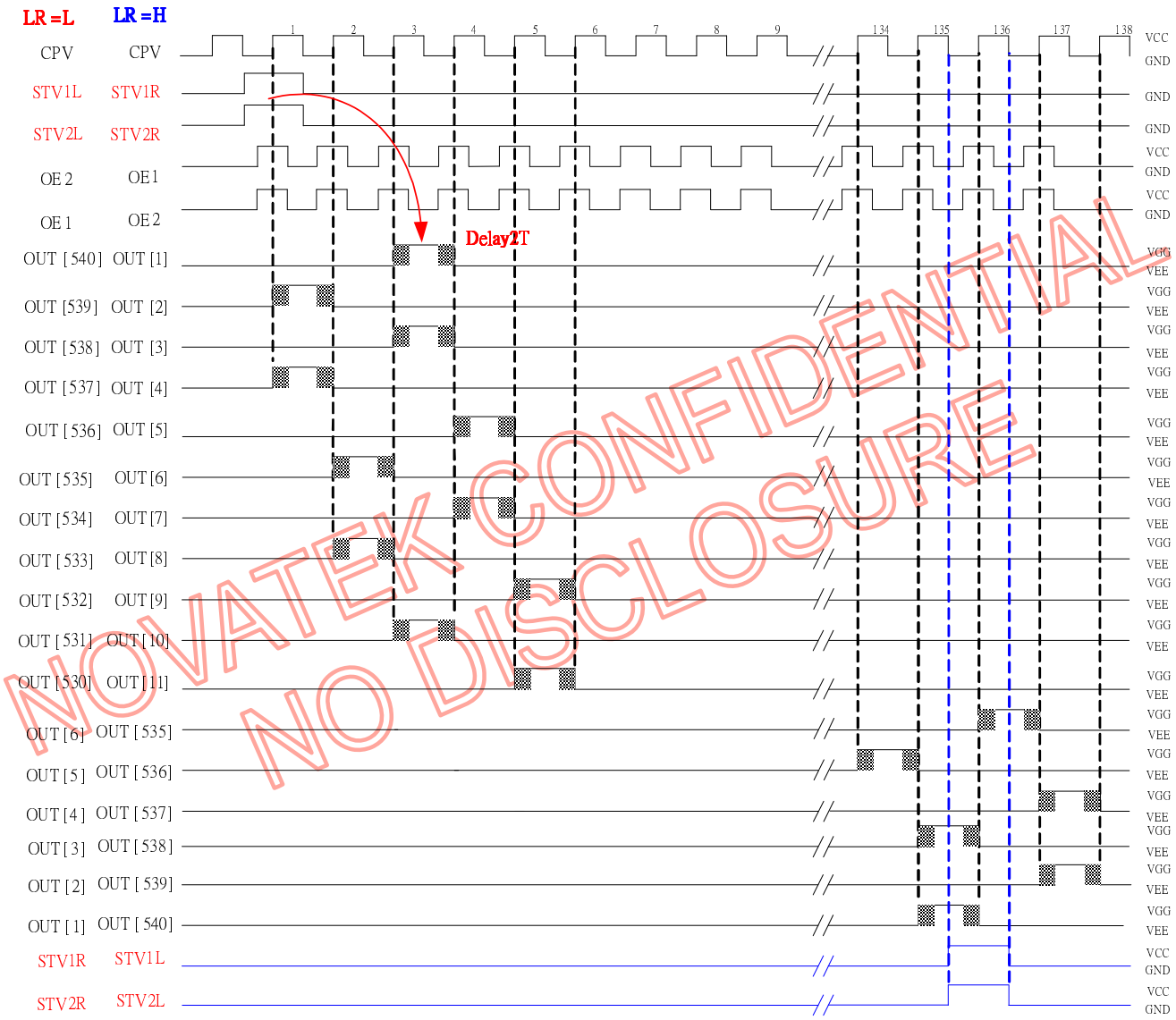
7. Reverse Jumping One pulse mode, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	L	L	L

("H" level =VCC, "L" level =GND)



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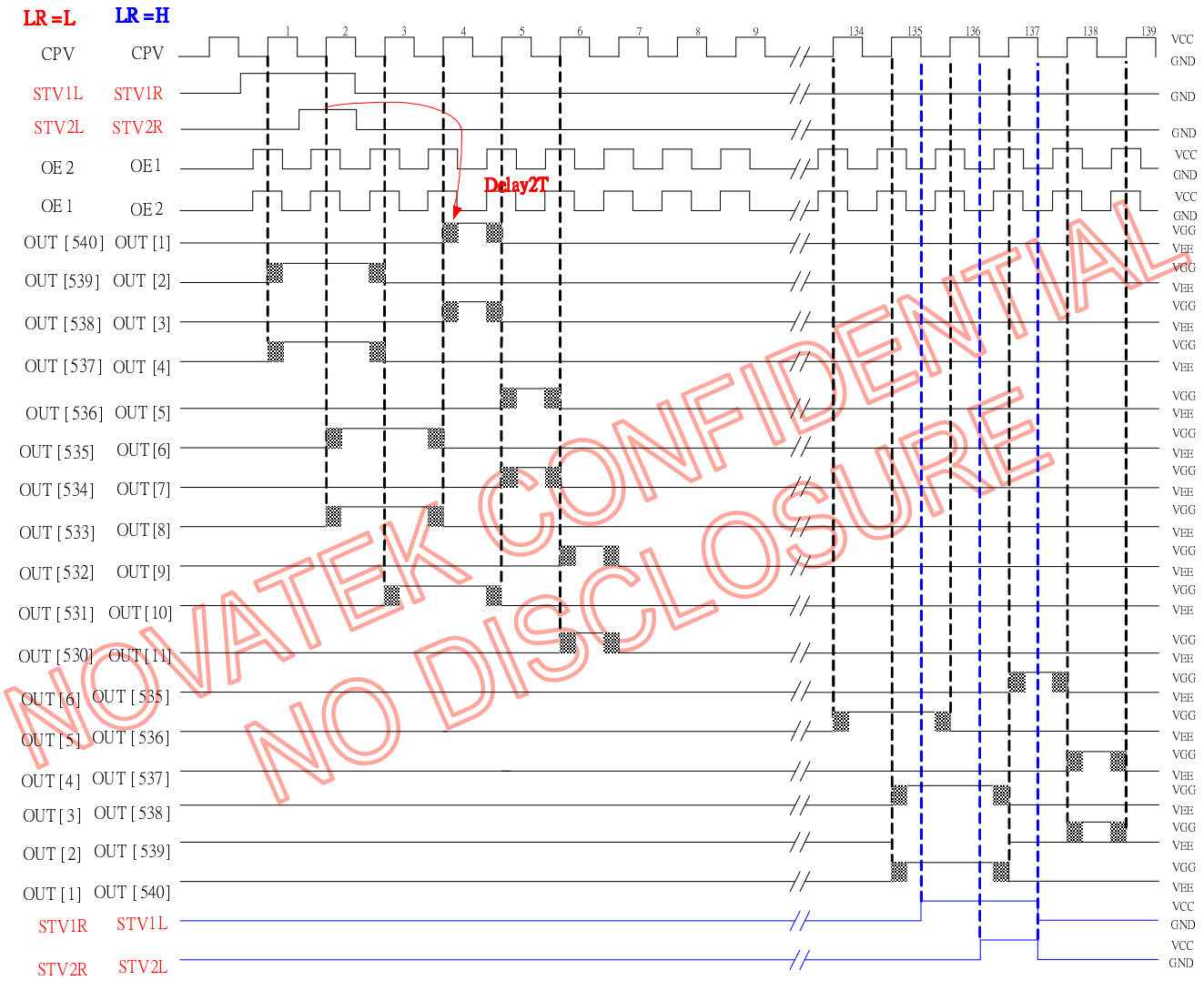
8. **Reverse Jumping One pulse mode (Long STV)**, Output channel = 540

LR	MODE1	MODE2	CHIP_SEL2	CHIP_SEL1	SEL	UD	OEPSN
H/L	H	H	H	H	L	L	H

("H" level =VCC, "L" level =GND)



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Absolute Maximum Rating*

Logic supply voltage, VCC	-0.3V to 5.0V
Supply voltage, VEE	-20V to +0.3V
Supply range, VGG-VEE	-0.3V to 45V
Supply voltage, VGG	-0.3V to 40V
Operating temperature	-30℃ to 85℃
Storage temperature	-55℃ to 125℃

*Comments
Stresses above those listed under "Absolute Maximum Rating" may cause permanent damage to the device. These are stress ratings only. Functional operation of this device at these or any other conditions above those indicated in the operational sections of this specification is not implied and exposure to absolute maximum rating conditions for extended periods may affect device reliability.

6. DC Electrical Characteristics

Symbol	Parameters	Min.	Typ.	Max.	Unit	Conditions
VGG	VGG Voltage	VCC	30	VEE+40	V	
VEE	VEE Voltage	-15	-	-5	V	
Vxo	Voltage Range of VGG – VEE	12	-	40	V	VCC = 3.3V
VCC	VCC Supplied Voltage	2.3	3.3	3.6	V	
VIH	High Level Input Voltage	0.7*VCC	-	VCC	V	VCC = 3.3V
VIL	Low Level Input Voltage	0	-	0.3*VCC	V	VCC = 3.3V
IXOH	High Level Output Current	0.5	-	-	mA	Driving current, VO = VGG – 0.5V, VGG –VEE = 30V
IXOL	Low Level Output Current	-0.5	-	-	mA	Sink current, VO = VEE + 0.5V, VGG –VEE = 30V
IPOH	High Level Output Current	200	-	-	uA	STV1/STV2, VO = VCC – 0.3V
IPOL	Low Level Output Current	-200	-	-	uA	STV1/STV2, VO = 0.3V
Rin	Pull-up & Pull down Impedance	70K	200K	-	Ω	Note1
IIL	Input Leakage Current	-	-	±1	uA	Note2
IDD	Operating Current Consumption	-	15	100	uA	VCC=3.3V, F_{CPV} =50KHz, No load Note3
IGG	Operating Current Consumption	-	50	250	uA	VGG=25V, F_{CPV}=50KHz, No load Note3
IEE	Operating Current Consumption	-	-50	-250	uA	VEE=-10V, F_{CPV} =50KHz, No load Note3

(VGG=25V, VEE= -15V, VCC=3.3V, GND=0V, TA= 25℃)
Note1 : XAO_L/R, SEL_L/R, UD_L/R, MODE1_L/R, MODE2_L/R, CHIP_SEL2_L/R, OEPSN, STV_MODE_L/R
Note2 : Except --- XAO_L/R, SEL_L/R, UD_L/R, MODE1_L/R, MODE2_L/R, CHIP_SEL1_L/R, CHIP_SEL2_L/R, OEPSN, STV_MODE_L/R
Note3 : For only one output channel turns on during one CPV period

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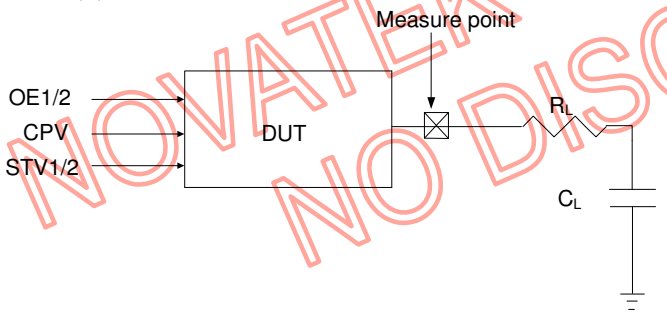
7. AC Electrical Characteristics

Symbol	Parameters	Min.	Typ.	Max.	Unit	Conditions
Tdt	STV1/STV2 Delay Time	-	-	550	ns	C_L = 20pF
Tdo	Driver Output Delay Time	-	-	300	ns	C_L = 300pF, R_L = 5KΩ Note(1)
Tthl	Output Falling Time	-	-	400	ns	C_L = 300pF R_L = 5KΩ Note(1), 90% to 10%
Ttlh	Output Rise Time	-	-	500	ns	C_L = 300pF, R_L = 5KΩ Note(1), 10% to 90%
TXAO	XAO_L&XAO_R to Driver Output Delay Time	-	-	20	us	C_L = 300pF
Toe	OE1_L, OE1_R, OE2_L, OE2_R to Driver Output Delay Time	-	-	300	ns	C_L = 300pF, R_L = 5KΩ Note(1),
F_{CPV}	Clock Frequency	-	-	200	kHz	In cascade connection
Trck	Clock Rise Time	-	-	100	ns	C_L = 20pF
Tfck	Clock Falling Time	-	-	100	ns	C_L = 20pF
PWCPV	Clock Pulse Width (High & Low)	500	-	-	ns	
Tsu	STV1/STV2 Set-up Time	200	-	-	ns	
Thd	STV1/STV2 Hold Time	300	-	-	ns	
Twcl	Output Enabled pulse width	0.5	-	-	us	

(VGG=25V, VEE= -15V, VCC=3.3V, GND=0V, TA= 25℃)

Note (1)



C_L = 300pF , R_L = 5kΩ

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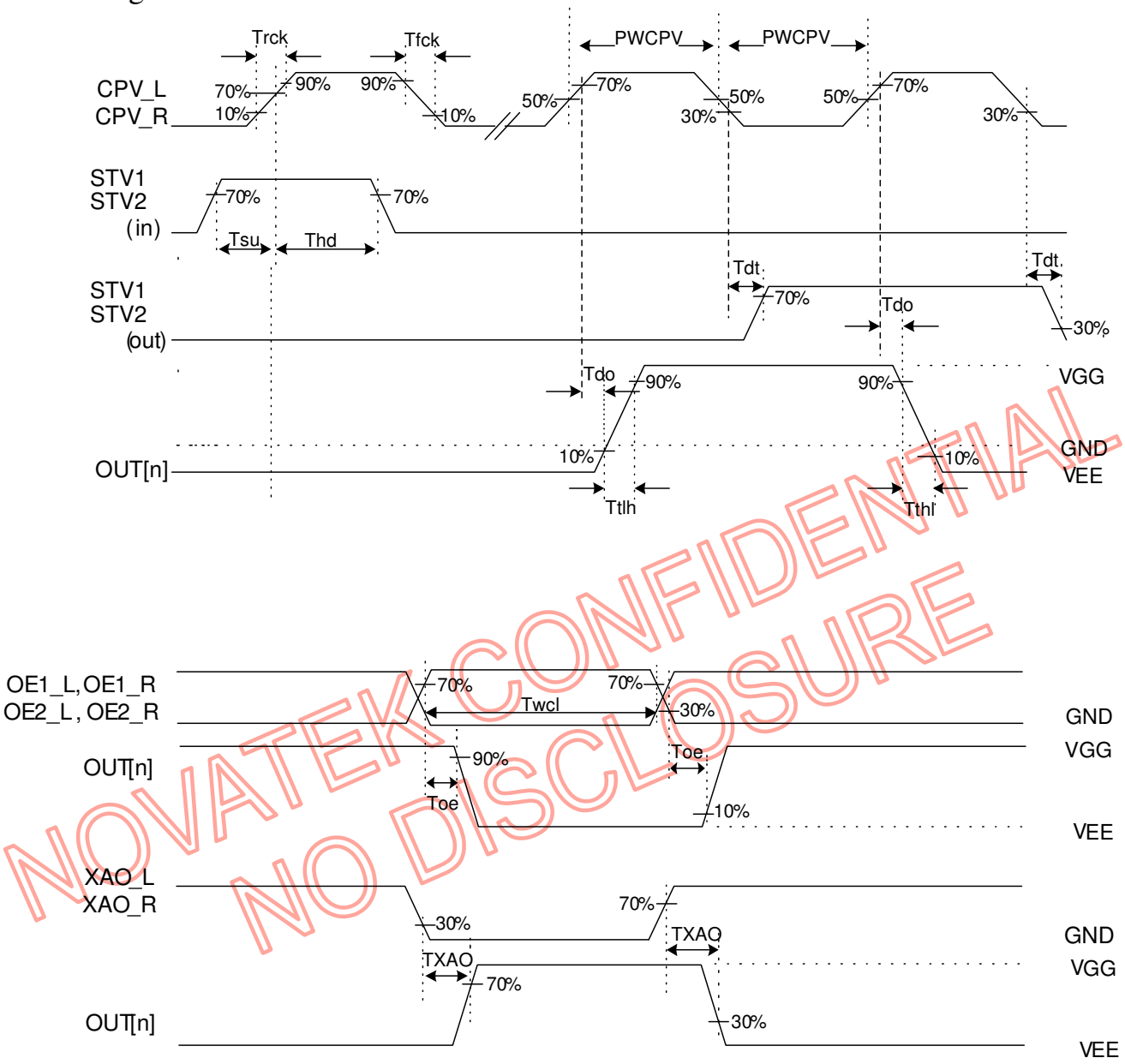
8. Timing Waveform



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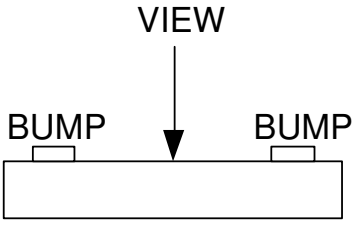
9. Pin Assignment



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10. Revision History

NT39565D Specification Revision History			
Version	**Content**	**Page**	**Date**
1.0	Original	All	2012/04/25

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